

Seminar Report

Machine Learning - Assignment 04

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2 Gaussian Mixture Models

2.1 Studying and Clustering the Toy GMM Dataset

In the first part of the task, we analyzed the provided GMM generator. For this reason, we generated the `toy_gmm` dataset, as shown in [1](#). It shows five gaussian distributions with different means, variances and sizes, in the two dimensional space.

Next, we tried to cluster the dataset with K-Means using five components, see Figure [2](#). Clearly, K-Means is not capable to separate all of the five gaussian distributions. Two of the gaussians that are a bit more separated from the other gaussians are correctly identified as two separate classes. For the remaining three gaussians, K-Means applies seemingly arbitrary cuts. Here, the weaknesses of K-Means become apparent, namely that it struggles with

- clusters of different sizes
- elongated clusters
- overlapping clusters.

2.2 Fitting a GMM Model and Comparison with K-Means

In order to obtain a better clustering, we implemented an EM-algorithm to fit a GMM model to the dataset.

One can see that the GMM clustering approach that utilizes the EM algorithm works much better for the problem than K-Means clustering, which makes sense as the data was generated using a GMM model as well. All the cluster assignment by the GMM model seem correct, compared to the K-Means model showing the same problems as explained in the previous section [2.1](#).

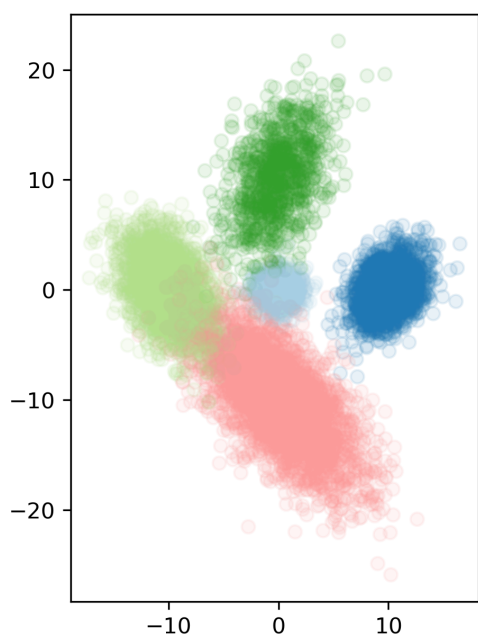


Figure 1: The Toy GMM Dataset

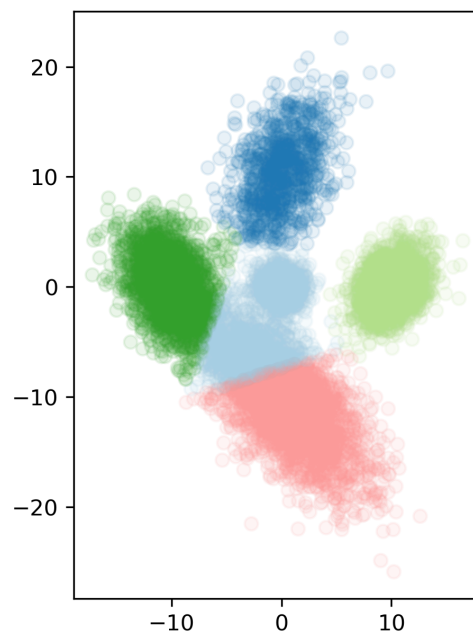


Figure 2: K-Means Clustering applied to the Toy GMM Dataset

The next two plots show the clustering results and comparison using four (4) and six (5) components, respectively. For four clusters, K-Means seems to perform even better than for five components, merging gaussians into one cluster and correctly separating the remaining gaussians, but this might as well be a lucky shot. The gmm model merges the two larger clusters, clearly underfitting the original distribution.

For six clusters, the gmm model still exhibits only five clusters. This is probably because the gmm model is capable to model the sizes of the clusters as well. It seems like the model has assigned a size of nearly 0 to the sixth cluster such that it is not visible in the plot and the result still matches the optimal clustering result using five components. In contrast, K-Means has no way to model the sizes of the clusters, and hence introduces a sixth cluster separating points that clearly belong to the same gaussian.

To sum it up, for the gmm model is seems important to specify the correct amount of clusters or more than that such that the model can decide itself which sizes to assign to the clusters. With too few clusters specified, gmm is not capable to correctly cluster the points. As we know from lecture, in general K-Means is even more sensitive to a proper choice of the amount of clusters. In this task, however, K-Means fail for all $n \in \{4, 5, 6\}$ because of the aforementioned reasons.

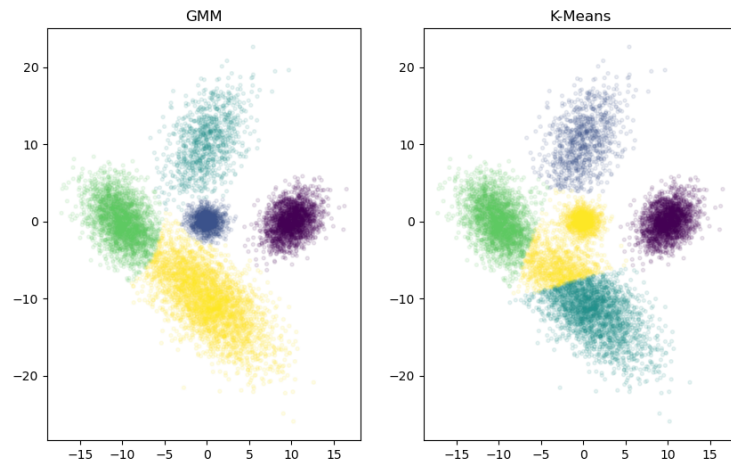


Figure 3: Comparison: GMM clustering vs. K-Means clustering using 5 components

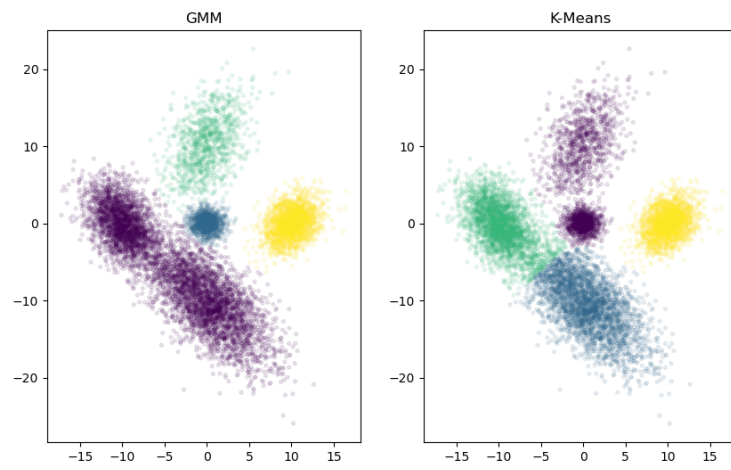


Figure 4: Comparison: GMM clustering vs. K-Means clustering using 4 components

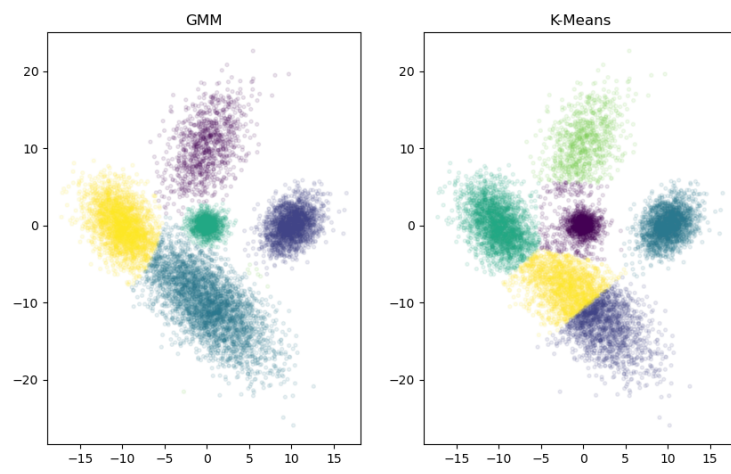


Figure 5: Comparison: GMM clustering vs. K-Means clustering using 6 components

Ehrenwörtliche Erklärung

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Declaration of Used AI Tools

Task	Solved with AI?	Tool	Purpose	Useful?
2c)	yes	gemini	debugging	partly

Unterschrift

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